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Part 1: Web Development & Architecture

Live Deployed URL: https://crystal-kizor-cyan.vercel.app/

Stack: React 19 • TypeScript • Tailwind CSS v4 • GSAP

The Crystal Kizor platform brings seven distinct ventures across architecture, furniture, media, and youth movements under one unified digital presence. Rather than presenting a disjointed list of businesses, the site is structured around her core philosophy: **climate-responsive intelligence, human potential, and African context**.

KEY TECHNICAL DECISIONS

- **Stack:** Built with React 19, TypeScript, Tailwind CSS v4, and GSAP. This combination delivers clean type safety, reliable animation timelines, and minimal CSS footprint.
- **Information Architecture:** Created an interactive four-tier taxonomy allowing visitors to filter initiatives by discipline, paired with a dedicated "Where Next" guided pathway section that pre-populates the inquiry router based on visitor persona.
- **Performance & Polish:** Features a custom GSAP architectural intro with `prefers-reduced-motion` compliance, WebP image compression with high-priority preloading, responsive scroll reveals, and zero layout shift.
- **Code Standards:** Strictly follows production-grade conventions: PascalCase components, 100% arrow functions, one-line JSDocs, and standard Tailwind utility tokens.

Part 2: AI Product Thinking: BuildIQ by TEA

BuildIQ by TEA is an interactive mobile intelligence platform that deconstructs real-world African architectural projects into living, technical case studies. It is designed for African architecture graduates, students, and emerging practitioners seeking practical design and construction mastery.

PROBLEM SOLVED

University curricula across Africa overwhelmingly prioritize abstract Western theory, leaving graduates unprepared for erratic power grids, humid microclimates, and volatile material costs. Crucially, the actual design decisions, thermodynamic physics, and material compromises behind successful regional buildings remain locked inside senior firms' private archives.

HOW IT IS USED

Users explore a catalog of building typologies. Selecting a project reveals high-resolution assembly photographs with interactive callout lines and clickable section hotspots, or a mobile-optimized 3D spatial view when CAD or blueprints are available. Tapping an overhang, wall, or courtyard displays its material composition, thermal mass math, airflow physics, and cost trade-offs. An embedded AI assistant allows architects to query the project's decision history and brainstorm modifications for their own active briefs.

TECHNOLOGY & AI MODEL

- **Model:** Gemini 1.5 Flash via Google Cloud Vertex AI, selected for its multimodal vision to spatially ground hotspots on drawings/photos and low-latency mobile inference.
- **Architecture:** React Native mobile client with Three.js for 3D glTF models; a vector database (`pgvector`) indexing on-site architect interview transcripts, climate data (Open-Meteo), and technical drawings.

MVP BUILD (3 WEEKS)

TEA's field team conducts on-site guided walkthroughs across 5 flagship Nigerian projects (including Studio COKA's Nsukka off-grid hospital and Project ARAM). We record architect interviews, capture key assembly photos, map interactive hotspots, and connect a RAG-backed Gemini chat interface.

LIMITATIONS & SAFEGUARDS

- **Limitation:** Case studies provide educational guidance, not certified structural or MEP engineering calculations.
- **Safeguards:** Prominent disclaimer requiring licensed engineer sign-off; outputs are strictly grounded in verified on-site project documentation to eliminate hallucinations.

Part 3: Analytics & Improvement

PART 3A: MEASUREMENT FRAMEWORK

To measure website health, I track three metric layers:

1. **Performance & Engagement:** Page load speed and Core Web Vitals (PageSpeed), bounce rate, and time spent on page (Google Analytics 4).
2. **Discovery & Acquisition:** Organic search clicks and impressions (Google Search Console), plus referral channels via UTM parameters.
3. **UX & Attention:** Dwell time per section and scroll depth heatmaps (Microsoft Clarity) to identify where visitors focus versus what they skip.

Iterative Improvement: Re-order layouts so high-dwell sections appear earlier, rewrite content where reading time is abnormally low, and optimize heavy assets if mobile load speed lags.

PART 3B: SCENARIO INVESTIGATION (5,000 VISITORS, 5 ENQUIRIES)

A 0.1% conversion rate (5/5,000) indicates problems extending beyond technical execution into marketing, sales positioning, and visual psychology:

What to Investigate:

1. **Technical Health:** Verify mobile load speed, device responsiveness, and test form functionality across browsers.
2. **Marketing & Sales Positioning:** Evaluate bounce rates per UTM source. Is the value proposition clear within five seconds, or does dense grammar, industry jargon, and weak positioning confuse visitors?
3. **Visual Attention & CTAs:** Use heatmaps to analyze visual hierarchy: are images drawing the eye toward the offer, or are poorly placed CTAs being ignored?

Next Steps:

1. Fix technical bottlenecks and ensure sub-2s mobile load times.
2. Refine copy into clear, persuasive, benefit-driven language that sharpens the sales hook.
3. Redesign visual flow using directional imagery to guide attention to prominent, well-placed CTAs, while simplifying form fields to bare essentials.